

UIJONG YANG

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SUMMARY

- Backend and cloud software engineer who pins down root causes through measurement and improves on that evidence.
- Cut core API latency by up to **42%** at Boeing Korea and built **750+ automated tests (76% coverage)** from scratch at PTKorea.
- Active contributor to large open-source projects, including Uber Cadence and vLLM Semantic Router.

SKILLS

Languages	Python, Go, C#, SQL, Rust, Java
Backend	FastAPI, ASP.NET, Django, Spring Boot
Cloud / Infra	AWS, Terraform, CloudFormation, Docker
Testing / CI	Pytest, Go testing, GitHub Actions

CERTIFICATIONS

AWS Advanced Networking (ANS), AWS Solutions Architect Associate (SAA), CCNA (Expired), TOEIC Speaking IH

EXPERIENCE

Boeing Korea, Cloud System Software Engineer Intern Oct 2025 ~ Mar 2026
C#, Python, ASP.NET, EF, SQL Server, Docker | Seoul, Korea

- **Core API Optimization:** Isolated unavoidable latency with ping tests between the US database and the Korea office, profiled EF-generated SQL with SET STATISTICS, then converted subqueries to JOIN/EXISTS, added composite indexes, and cut DB round-trips. **Table-view query 42% (2046ms -> 1189ms), single-record detail query 41% (1389ms -> 823ms).**
- **Graph UI Rendering:** Slimmed server-side DTOs and moved Dagre data-structure transforms to the server so the browser only renders. **Resolved rendering that took 5+ minutes (10+ on slower internal networks), ~20% faster** by JavaScript-based measurement.
- **AI Model CPU Fallback:** Traced the call stack of a GPU(CUDA)-only model and reimplemented an equivalent CPU-based Python fallback. **Removed GPU dependency** and deployed into CPU-only production and development environments.
- Visualized complex node dependencies via Dagre topological sorting, and rebuilt the Dockerfile with an internally certified base image to support an international demo session.

PTKorea, Python Developer Intern Jun 2025 ~ Sep 2025
FastAPI, Pytest, PostgreSQL, OpenCV, PaddleOCR | Seoul, Korea

- **Test Automation:** In a codebase with no tests, traced every function's call stack into a dependency diagram and built unit/integration tests bottom-up from the lowest-dependency modules. **750+ tests, 76% coverage from zero**, blocking regressions and improving release stability.
- Maintained the FastAPI backend with bug fixes, and compared and analyzed OpenCV/Pillow image-processing performance.

System Software Laboratory, Soongsil University, Backend AI Integration Assistant Sep 2024 ~ Jun 2025
FastAPI, Celery | Seoul, Korea

- **Backend Development:** Built a FastAPI backend for a font-generation web application, integrating asynchronous AI workflows with Celery.
- **AI Integration:** Integrated multiple font-generation AI models with the internal backend and standardized the APIs for model execution.

PROJECTS

Single Sign-On Service (ITSUPPORT-AUTH) Ongoing
Go, AWS Cognito, Lambda, API Gateway, SQS | Soongsil University, ITSUPPORT Committee

- Built an OIDC-based SSO on AWS serverless for multiple internal services; an API Handler proxies Cognito (login, logout, onboarding, token exchange/refresh) and centralizes rate limiting, CORS, and JWT-protected routes.
- **Async event pipeline:** Decoupled re-login synchronization as Post Authentication -> SQS -> Consumer (failures isolated to a DLQ), guaranteeing idempotency with SQL transactions and unique constraints. **Login response 637ms -> 176ms (~72% faster, post cold-start).**
- **Cold-start mitigation:** Rewrote the auth system from Python to Go and applied Provisioned Concurrency. **~124ms average API Handler latency**, sustained with no reported service interruptions during a two-week scholarship application period (Grafana Cloud).

AWS Infrastructure Management (ITSUPPORT-AWS)

Ongoing

Terraform, AWS, CloudFormation, GitHub Actions | Soongsil University, ITSupport Committee

- Designed and provisioned a 3-tier AWS infrastructure as code with Terraform and CloudFormation, integrating pre-commit hooks (checkov, tfint) to enforce deployment security and code quality.
- **Cost optimization:** Consolidated dev instances into a single EC2 Spot instance, moved to ARM64 (Graviton), and removed idle EC2/RDS resources. **44.4% lower monthly cost (\$203.32 -> \$112.93).**
- **Security hardening:** Redesigned security groups with least privilege to block direct external DB access, replaced SSH key pairs with audited SSM-only access, and enforced IMDSv2.

RAG Agent System

Personal Project

Python, FastAPI, ChromaDB, gpt-4o-mini

- Optimized a Korean-law multiple-choice (KMMLU) RAG system across 6 phases and 33 experiments using a hypothesize -> implement -> measure -> log-analysis loop. **Baseline 51.58% -> 53.28% (dev set).**
- Identified counterintuitive, log-driven findings ("more context is not better", "generation tuning beats retrieval tuning", "the model's own reasoning was underrated") and documented them.

Her, On-Device LLM English Conversation App

Personal Project

Go, WebSocket, LiteRT-LM, Gemma 4

- Built a fully local conversation app that runs without cloud or external API keys. A Go orchestrator over WebSocket keeps per-connection session history and streams a local LLM (Gemma 4) token by token.
- Passed push-to-talk audio directly to the audio-native model, removing the STT step. Voice never leaves the device, preserving privacy.

OPEN SOURCE CONTRIBUTIONS

Cadence, Distributed Workflow Orchestration Engine (Uber)

Go

- Replaced a custom rate limiter with the Go standard library (rate.Limiter), reducing complexity (#7585); reproduced a permanent deadlock on negative RPS via a self-written POC and added a validation guard.
- Removed redundant synchronous retries in task matching (#7650), propagated context timeouts and structured wait times (#7709), handled timeouts with select plus async re-matching (#7827), and added dynamic config for the Adaptive TaskList Scaler (#8035). **5 PRs merged.**

vLLM Semantic Router, LLM Router

Go

- Refactored Redis/Milvus cache configuration into a consistent structure, with a fallback path for legacy settings and unit tests (#1100).

Youki, Container Runtime

Rust

- Fixed bugs and added unit tests (#3602, #3608). Analyzed why tty inode validation defends against CVE-2025-52565 (TOCTOU), down to fstat/fstatfs behavior, and wrote a technical article before contributing.

EZ, Programming Language

- Added HTTP status code constants and fixed a bug in the 'using' keyword (#907, #979).

EDUCATION

Soongsil University, B.S. Computer Science

Mar 2022 ~ Feb 2027

GPA 3.79 / 4.5 | All graduation requirements completed